

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

Technical Presentation

<i>Criteria</i>	<i>Technical Content Accuracy(3)</i>	<i>Problem Identification(2.5)</i>	<i>Use of Tools(2)</i>	<i>Communication(1.5)</i>	<i>Professionalism(1)</i>	<i>Total(10)</i>
<i>Weightage</i>	30%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						

Code of Conduct:

I A. LOKESH KUMAR (192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

Focus Areas:

- Definition and need of DSS
 - Components (Data, Model, UI)
 - Types of DSS
 - Real-world applications (banking, healthcare)
 - Advantages & limitations
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Topic 2: Operational Systems vs DSS

Focus Areas:

1. OLTP vs OLAP concepts
 2. Key differences (data, users, purpose)
 3. Performance comparison
 4. Use-case scenarios
 5. Integration between systems
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Topic 3: Architecture of Data Warehouse

Focus Areas:

1. Data warehouse definition
 2. Three-tier architecture
 3. ETL process
 4. Data marts & metadata
 5. Benefits in decision-making
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Topic 4: OLAP and Data Cubes

Focus Areas:

1. OLAP concept
 2. Data cube structure
 3. OLAP operations (slice, dice, roll-up, drill-down)
 4. Applications in analytics
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Topic 5: Dimensional Data Modeling

Focus Areas:

1. Fact and dimension tables
 2. Star schema vs snowflake schema
 3. Schema design process
 4. Real-world example
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Topic 6: Query Processing & BI Tools

Focus Areas:

1. Query processing stages
 2. Query optimization
 3. BI tools overview
 4. Reporting and visualization
 5. Examples of open-source BI tools
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Topic 7: OLAP Application using KNIME

Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation
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PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
 2. Each member must participate
 3. Assign roles:
 1. Presenter
 2. Designer (PPT)
 3. Researcher
 4. Demo/Case handler
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◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
 1. Definition and importance
 2. Core Concepts
 1. Explanation with diagrams
 3. Process/Architecture
 1. Step-by-step explanation
 4. Real-World Example
 1. Case study or application
 5. Tool/Demo (if applicable)
 1. KNIME or BI tool
 6. Conclusion
 1. Summary + future scope
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◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
 1. Diagrams

2. Flowcharts
 3. Tables (for comparison)
 3. Avoid:
 1. Too much text
 2. Copy-paste content
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◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
 1. Eye contact
 2. Clear voice
 3. Logical flow



Engineer to Excel

SIMATS

ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

FROM RAW DATA TO SMART DECISIONS

"Transforming raw data into meaningful insights for smarter business decisions."

GUIDE:
BEULAHDAVID. D

Presented By:
A. LOKESH KUMAR (192524157)

WHY DO WE NEED A DATA WAREHOUSE?

Imagine a Supermarket

Every day, it generates data from:

 Sales Transactions

 Inventory

 Customers

 Suppliers

 Payments

Challenge

Data is scattered across different systems.

Difficult to analyze and generate reports.

Slow decision-making.

Solution: Data Warehouse

A **Data Warehouse** is a centralized repository that:

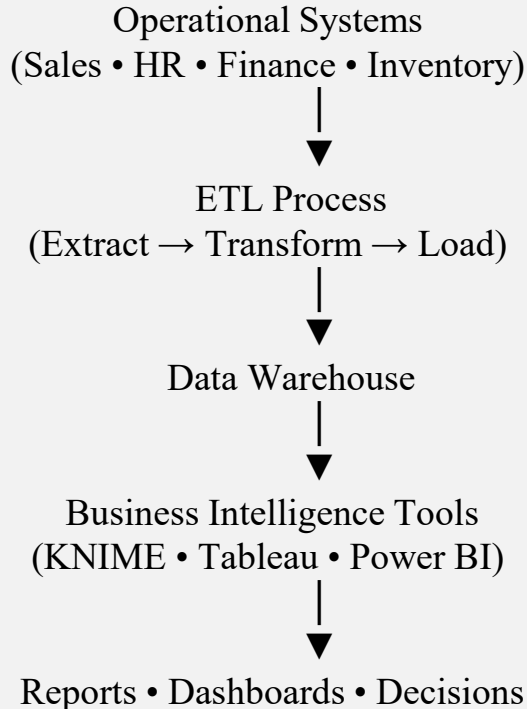
Collects data from multiple sources.

Stores historical data.

Supports analysis and reporting.

DATA WAREHOUSE ARCHITECTURE & ETL PROCESS

DATA WAREHOUSE ARCHITECTURE



ETL Process

Extract

- Collects data from multiple sources.

Transform

- Cleans, validates, and converts data into a standard format.

Load

- Stores the processed data in the Data Warehouse.

Advantages

-  Integrates data from multiple sources
-  Improves data quality
-  Enables fast reporting and analytics
-  Supports better decision-making

OLAP & DATA CUBES

What is OLAP?

OLAP (Online Analytical Processing) is a technology used to analyze large volumes of data from multiple dimensions, helping businesses make faster and smarter decisions.

Data Cube Dimensions

- 📅 **Time** (Day, Month, Year)
- 📦 **Product**
- 🌍 **Location**

OLAP Operations

- **Slice** – View data for one specific dimension.
- **Dice** – Analyze a selected subset of data.
- **Roll-up** – Summarize data (Day → Month → Year).
- **Drill-down** – View detailed data (Year → Month → Day).

Benefits of OLAP

- ⚡ Fast data analysis
- 📊 Interactive reports
- 📈 Trend and performance analysis
- 🎯 Better business decisions

DIMENSIONAL MODELING & BUSINESS INTELLIGENCE

Dimensional Modeling

A technique used to organize data for **easy analysis and faster query performance**.

Star Schema ★

- Simple structure
- Central **Fact Table** connected to **Dimension Tables**
- Faster query execution
- Easy to understand

Snowflake Schema ❄️

- Normalized dimension tables
- More complex structure
- Reduces data redundancy
- Better data integrity

Business Intelligence (BI) Tools

BI tools convert warehouse data into meaningful reports and dashboards.

-  **KNIME** – Data analytics & workflows
-  **Tableau** – Interactive visualizations
-  **Power BI** – Business dashboards

Real-World Applications of Data Warehousing



Retail (Amazon, Walmart)

- Sales analysis
- Inventory management
- Customer purchase trends



Banking

- Fraud detection
- Risk analysis
- Customer transaction monitoring



Healthcare

- Patient record management
- Disease trend analysis
- Hospital performance reports



Entertainment (Netflix, Spotify)

- Personalized recommendations
- User behavior analysis
- Content performance tracking

Thank you!

